

实验结果

数据集

TPC-H 的 `lineitem` 表子集

每一行是一条订单中的商品明细记录，字段用 `|` 分隔，没有表头。

字段结构

TPC-H `lineitem` 标准字段顺序是：

代码块

```
1  0  l_orderkey
2  1  l_partkey
3  2  l_suppkey
4  3  l_linenumber
5  4  l_quantity
6  5  l_extendedprice
7  6  l_discount
8  7  l_tax
9  8  l_returnflag
10 9  l_linestatus
11 10 l_shipdate
12 11 l_commitdate
13 12 l_receiptdate
14 13 l_shipinstruct
15 14 l_shipmode
16 15 l_comment
```

例如第一行：

代码块

```
1  1|3104|170|1|17|17120.70|0.04|0.02|N|0|1996-03-13|...
```

当前 M-HIBE 实验用了哪些字段

二维版本只用两个维度：

代码块

```
1  Coords[0] = l_shipdate
```

```
2  Coords[1] = l_discount * 100
```

也就是：

代码块

```
1  X 维：发货日期 shipdate
2  Y 维：折扣 discount
```

代码里：

代码块

```
1  p.Coords[0] = ParseDate(cols[10])
2  dFloat, _ := strconv.ParseFloat(cols[6], 64)
3  p.Coords[1] = int64(dFloat * 100)
```

`ParseDate` 会把日期转成从 `1992-01-01` 起算的天数。

例如：

代码块

```
1  1994-01-01 -> 某个整数 day offset
2  1994-12-31 -> 另一个整数 day offset
```

`discount` 原来是小数：

代码块

```
1  0.05, 0.06, 0.07
```

乘以 100 后变成整数：

代码块

```
1  5, 6, 7
```

这样方便映射成二进制前缀。

当前查询条件

代码里的查询是 TPC-H Q6 风格的一部分：

```
代码块
1 shipdate in [1994-01-01, 1994-12-31]
2 discount in [5, 7]
```

也就是：

代码块

```
1 1994 年全年
2 折扣 0.05 到 0.07
```

当前输出里：

代码块

```
1 Query matched 5139 real records
```

表示有 5139 行满足这个二维范围查询。

但这些记录会落到更少的唯一二维坐标上：

代码块

```
1 5139 matching rows collapsed into 1085 unique spatial points
```

原因是很多行可能有相同的：

代码块

```
1 (shipdate, discount)
```

例如多条订单明细都在同一天发货，并且折扣都是 0.06。

为什么查询空间是 1095 个点

查询范围是：

代码块

```
1 365 天 × 3 个折扣值
```

即：

代码块

```
1  365 × {5,6,7} = 1095 个二维点
```

实际数据占了：

代码块

```
1  1085 个唯一点
```

所以空点是：

代码块

```
1  1095 - 1085 = 10
```

这正好对应你的输出：

代码块

```
1  Query-Scoped Empty Regions: 10
2  10 empty-region keys cover 10 empty spatial points
```

这个数据集在实验里的作用

它主要用于模拟一个外包数据库范围查询场景：

代码块

- 1 客户端查询某个多维范围
- 2 服务器返回命中的记录
- 3 同时给出证明：
 - 4 1. 返回的记录确实来自数据库
 - 5 2. 查询范围内没有遗漏其他记录

在当前代码里：

- `lineitem_120K.tbl` 提供真实数据库点。
- `M-HIBE / WKD-IBE` 用来证明查询范围内的空区域。
- `ZK accumulator` 用来证明返回记录属于数据库。
- 空区域 key 用来证明完整性，也就是“没有漏掉查询范围内的数据”。

[*] Global Setup Time: 4691.54 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 38.13 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/delegation
5. Prefix Extraction Time: 3.89 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB
9. WKD-IBE Delegation Time: 251.45 ms

-> Engine A Total: 255.34 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 251.2196 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 24215.68 ms

[+] ZK Prover Time: 2657.61 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.60 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 27128.63 ms (27.13 s)

Total Client Verification Time: 253.82 ms

复杂度

符号定义

设：

代码块

- 1 N = 数据库记录数
- 2 M = 数据库中唯一多维点数
- 3 d = 维度数
- 4 l = 每维 bit 长度
- 5 $U = 2^l$, 每维域大小
- 6 Q = 查询范围
- 7 K = 查询返回记录行数
- 8 mQ = 查询范围内唯一真实点数
- 9 VQ = 查询范围体积
- 10 $EQ = VQ - mQ$, 查询范围内空点数
- 11 Cq = 查询范围 canonical cover 数
- 12 Gq = 查询触达的全局空父区域数
- 13 Sq = 最终 query-scoped empty key 数

当前实验里：

代码块

```
1  N  = 120515
2  d  = 2
3  l  = 12
4  K  = 5139
5  mQ = 1085
6  VQ = 1095
7  EQ = 10
8  Cq = 12
9  Gq = 60
10 Sq = 10
```

离线复杂度

全库空区域索引构建：

代码块

```
1  Time:  $O(N \cdot l)$ 
2  Space:  $O(M \cdot l)$ 
```

原因是每个数据点的 X 坐标会插入到所有 X 前缀节点：

代码块

```
1  "", "0", "01", ..., full x
```

一共有 $l+1$ 个前缀。

如果离线真正物化所有全局空区域父 key，复杂度大致是：

代码块

```
1  Time:  $O(Gall \cdot \text{KeyGen})$ 
2  Space:  $O(Gall \cdot |SK|)$ 
```

其中：

代码块

```
1  Gall = 全库全局空父区域数量
2  |SK| = 一个 WKD-IBE secret key 大小
```

更细一点，生成全局空区域父节点前，需要对每个 X 前缀节点下的 Y 占用集合求空 canonical cover：

```
代码块
1 Time:  $O(\sum_v |Y_v| \log |Y_v| + Gall)$ 
2 Space:  $O(\sum_v |Y_v| + Gall)$ 
```

其中 v 是 X 前缀节点， Y_v 是该 X 前缀下出现过的 Y 值集合。

当前实验里没有完整物化 `Gall`，只构建了索引，并在查询时懒加载触达的父 key，所以输出里写的是：

```
代码块
1 Global Empty Parent Key Material Touched
```

不是全库总 key material。

在线查询复杂度

查询 canonical cover：

```
代码块
1 Time:  $O(d \cdot l + Cq)$ 
2 Space:  $O(Cq)$ 
```

二维时：

```
代码块
1  $Cq = Cx \cdot Cy$ 
```

每个一维区间 canonical cover 通常是 $O(l)$ 。

查找触达的全局空父区域：

```
代码块
1 Time:  $O(Tx + Gq)$ 
2 Space:  $O(Gq)$ 
```

其中 Tx 是与查询 X 范围相交的 X 前缀节点数量。

把全局空父区域裁剪到查询范围：

```
代码块
1 Time:  $O(Gq \cdot d \cdot l + Sq)$ 
```



```
2   Space:  $O(Sq)$ 
```

WKD-IBE 派生：

代码块

```
1   Time:   $O(Pq \cdot \text{KeyGen} + Sq \cdot \text{Delegate})$   
2   Space:  $O(Pq \cdot |SK| + Sq \cdot |SK|)$ 
```

其中：

代码块

```
1   Pq = 本次查询实际触达的全局空父 key 数  
2   Sq = 派生出的 query-scoped empty key 数
```

当前实验中：

代码块

```
1   Gq = 60  
2   Sq = 10
```

所以在线 M-HIBE 部分很小。

ZK accumulator 复杂度

设返回结果行数是 K ，补集大小是 $N-K$ 。

当前成员证明大致可以按：

代码块

```
1   Commitment:  $O(N)$   
2   Prover:       $O(K \log K)$  或  $O(K)$  到  $O(K \log K)$ ，取决于 accumulator/poly 实现  
3   Verifier:     $O(1)$  或接近  $O(1)$   
4   Proof size:  $O(1)$ 
```

从实验结果看：

代码块

```
1   ZK Prover Time  $\approx 2.4s$   
2   Proof Size  $\approx 1.20$  KB
```

也就是说瓶颈主要在 ZK prover，不在 M-HIBE 空区域派生。

整体复杂度总结

主方案可以写成：

代码块

```

1  Offline:
2    Empty index build:       $O(N \cdot l)$ 
3    Empty parent key gen:   $O(G_{all} \cdot \text{KeyGen})$ 
4    Accumulator setup:     $O(N)$ 
5
6  Online:
7    Query cover:            $O(d \cdot l + C_q)$ 
8    Empty parent lookup:    $O(T_x + G_q)$ 
9    Query intersection:     $O(G_q \cdot d \cdot l + S_q)$ 
10   Empty key delegation:   $O(P_q \cdot \text{KeyGen} + S_q \cdot \text{Delegate})$ 
11   ZK proving:             $O(K \log K)$  / implementation-dependent
12   ZK verification:       $O(1)$ 

```

3D

[*] Global Setup Time: 4809.78 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 3678.05 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543
4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/delegation
5. Prefix Extraction Time: 10249.09 ms
6. Query-Range Key Material: 14.27 KB
7. Global Empty Parent Key Material Touched: 1737.30 KB
8. Query-Scoped Empty Key Material: 1553.73 KB
9. WKD-IBE Delegation Time: 77160.74 ms

-> Engine A Total: 87409.83 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 1875.7370 ms (SUCCESS! Complete key cover.)

-> [Detail] 2327 matching rows collapsed into 2229 unique spatial points.

-> [Detail] 4543 empty-region keys cover 24051 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 8.

-> [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 24996.95 ms

[+] ZK Prover Time: 2125.52 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 3.19 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 114532.30 ms (114.53 s)

Total Client Verification Time: 1878.93 ms

Ponegly

2D

代码块

```
1 cargo test --package halo2-experiments --lib --  
  sql::q6_2d_final_v1::tests::test_1 --exact --nocapture
```

running 1 test

Number of columns: 45

test sql::q6_2d_final_v1::tests::test_1 has been running for over 60 seconds

Time to generate params 56.191891755s

Number of columns: 45

Time to generate vk 18.28329056s

Number of columns: 45

Time to generate pk 14.849952182s

Number of columns: 45

Prover Time 266.985772244s

Verifier Time 42.274822ms

test sql::q6_2d_final_v1::tests::test_1 ... ok

test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 18 filtered out; finished in 387.08s

3D

代码块

```
1 export RUST_MIN_STACK=33554432  
2 cargo test --package halo2-experiments --lib --  
  sql::q6_final_v2::tests::test_1 --exact --nocapture
```

running 1 test

Number of columns: 56

test sql::q6_final_v2::tests::test_1 has been running for over 60 seconds

Time to generate params 57.872005707s

Number of columns: 56

Time to generate vk 19.728859245s

Number of columns: 56

Time to generate pk 26.194673179s

Number of columns: 56

Prover Time 458.075726445s

Verifier Time 50.84907ms

test sql::q6_final_v2::tests::test_1 ... ok

test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 18 filtered out; finished in 600.34s

Zk-acc

2D

[*] Global Setup Time: 4588.36 ms

[*] Data Load + Direct Partition Time: 156.74 ms

[*] Loaded 120515 real TPC-H rows.

[*] Query matched 5139 rows, collapsed into 1085 unique 2D points.

[*] Full completeness empty-set enumeration: 10 points in 0.04 ms

[*] Commitment Time (DB + DB\I): 23345.97 ms

=== PROOF A: ZK-Membership for Correctness (I subset DB) ===

Proof time: 2038.96 ms

Verify time: 2.91 ms

Proof size: 1.20 KB

Status: true

=== PROOF B: ZK-NonMembership for Full Completeness (E disjoint DB) ===

[*] E contains every query point not returned by I: |E|=10

[*] Real aggregation padding: $|E|$ 10 -> 16 non-members

Proof time: 68355.82 ms

Verify time: 2.84 ms

Proof size: 1.32 KB

Status: true

=== FINAL NATIVE ZK-ACCUMULATOR REPORT ===

Architecture: 2D Native ZK-Accumulator Full Correctness + Full Completeness Baseline

Query partition sanity $|\text{unique}(I)| + |E| == |Q|$: true

Correctness (I subset DB): true

Completeness (every point in $Q \setminus I$ is non-member of DB): true

Overall baseline status: true

Total server proving time: 93740.79 ms (93.74 s)

Total client verification time: 5.75 ms

Total proof size: 2.52 KB

3D

[*] Global Setup Time: 4738.44 ms

[*] Data Load + Direct Partition Time: 194.19 ms

[*] Loaded 120515 real TPC-H rows.

[*] Query matched 2335 rows, collapsed into 2235 unique 3D points.

[*] Full completeness empty-set enumeration: 24045 points in 13.99 ms

[*] Commitment Time ($DB + DB \setminus I$): 24287.35 ms

=== PROOF A: ZK-Membership for Correctness (I subset DB) ===

Proof time: 1889.55 ms

Verify time: 2.59 ms

Proof size: 1.20 KB

Status: true

=== PROOF B: ZK-NonMembership for Full Completeness (E disjoint DB) ===

[*] E contains every query point not returned by I: $|E|=24045$

[*] Real aggregation padding: $|E|$ 24045 -> 32768 non-members

并行

2D (20 works)

[*] Global Setup Time: 5147.91 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 40.69 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/parallel delegation
5. Prefix Extraction Time: 3.55 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB
9. WKD-IBE Delegation Time: 35.18 ms

-> Engine A Total: 38.73 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 37.1339 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22985.10 ms

[+] ZK Prover Time: 2463.37 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.57 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D Parallel M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 25487.20 ms (25.49 s)

Total Client Verification Time: 39.70 ms

2D (10 works)

[*] Global Setup Time: 4446.81 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 38.19 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/parallel delegation
5. Prefix Extraction Time: 3.39 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB
9. WKD-IBE Delegation Time: 44.46 ms

-> Engine A Total: 47.85 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 34.1438 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22809.32 ms

[+] ZK Prover Time: 2441.06 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.45 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D Parallel M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 25298.23 ms (25.30 s)

Total Client Verification Time: 36.60 ms

2D (16 works)

[*] Global Setup Time: 4697.05 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 38.84 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/parallel delegation
5. Prefix Extraction Time: 3.98 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB
9. WKD-IBE Delegation Time: 36.24 ms

-> Engine A Total: 40.22 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 37.1128 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 25051.37 ms

[+] ZK Prover Time: 2461.55 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.38 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D Parallel M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 27553.14 ms (27.55 s)

Total Client Verification Time: 39.50 ms

2D (30 works)

[*] Global Setup Time: 4568.22 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 38.65 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/parallel delegation
5. Prefix Extraction Time: 3.72 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB
9. WKD-IBE Delegation Time: 36.94 ms

-> Engine A Total: 40.67 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 32.0702 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22362.10 ms

[+] ZK Prover Time: 2396.20 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.86 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D Parallel M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 24798.97 ms (24.80 s)

Total Client Verification Time: 34.93 ms

2D (50 works)

[*] Global Setup Time: 4535.82 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 37.81 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/parallel delegation
5. Prefix Extraction Time: 3.73 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB

9. WKD-IBE Delegation Time: 36.03 ms

-> Engine A Total: 39.77 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 33.2114 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22661.43 ms

[+] ZK Prover Time: 2490.47 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.34 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D Parallel M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 25191.66 ms (25.19 s)

Total Client Verification Time: 35.55 ms

2D (100 works)

[*] Global Setup Time: 4380.36 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100)

[*] Loaded 120000 real TPC-H records.

[*] Query matched 5119 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 33.10 ms

[+] Indexed X-Prefix Nodes: 5049

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 2D M-HIBE X-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 12
2. Global Empty Parent Regions: 60
3. Query-Scoped Empty Regions: 10
4. Cover Strategy: offline global empty parents + online query intersection/parallel delegation
5. Prefix Extraction Time: 3.29 ms
6. Query-Range Key Material: 5.00 KB
7. Global Empty Parent Key Material Touched: 1.88 KB
8. Query-Scoped Empty Key Material: 1.88 KB
9. WKD-IBE Delegation Time: 37.69 ms

-> Engine A Total: 40.99 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 34.2427 ms (SUCCESS! Complete key cover.)

-> [Detail] 5119 matching rows collapsed into 1085 unique spatial points.

-> [Detail] 10 empty-region keys cover 10 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 1.

-> [Detail] 10 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 21590.30 ms

[+] ZK Prover Time: 2384.57 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.44 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 2D Parallel M-HIBE X-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 24015.86 ms (24.02 s)

Total Client Verification Time: 36.68 ms

3D (20 works)

[*] Global Setup Time: 4526.61 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 4930.82 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543
4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/parallel delegation

5. Prefix Extraction Time: 12043.72 ms
6. Query-Range Key Material: 14.27 KB
7. Global Empty Parent Key Material Touched: 1737.30 KB
8. Query-Scoped Empty Key Material: 1553.73 KB
9. WKD-IBE Delegation Time: 8251.91 ms

-> Engine A Total: 20295.63 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 182.8246 ms (SUCCESS! Complete key cover.)

-> [Detail] 2327 matching rows collapsed into 2229 unique spatial points.

-> [Detail] 4543 empty-region keys cover 24051 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 8.

-> [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22966.02 ms

[+] ZK Prover Time: 1910.08 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.45 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D Parallel M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 45171.73 ms (45.17 s)

Total Client Verification Time: 185.27 ms

3D (16 works)

[*] Global Setup Time: 4469.49 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 4222.50 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543
4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/parallel delegation
5. Prefix Extraction Time: 10965.08 ms
6. Query-Range Key Material: 14.27 KB
7. Global Empty Parent Key Material Touched: 1737.30 KB
8. Query-Scoped Empty Key Material: 1553.73 KB
9. WKD-IBE Delegation Time: 8296.72 ms

-> Engine A Total: 19261.80 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 171.6235 ms (SUCCESS! Complete key cover.)

-> [Detail] 2327 matching rows collapsed into 2229 unique spatial points.

-> [Detail] 4543 empty-region keys cover 24051 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 8.

-> [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22497.09 ms

[+] ZK Prover Time: 1944.56 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.49 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D Parallel M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 43703.45 ms (43.70 s)

Total Client Verification Time: 174.11 ms

3D (30 works)

[*] Global Setup Time: 4407.58 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 4644.89 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543
4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/parallel delegation
5. Prefix Extraction Time: 10826.66 ms
6. Query-Range Key Material: 14.27 KB
7. Global Empty Parent Key Material Touched: 1737.30 KB
8. Query-Scoped Empty Key Material: 1553.73 KB
9. WKD-IBE Delegation Time: 7587.72 ms

-> Engine A Total: 18414.39 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 156.2981 ms (SUCCESS! Complete key cover.)

-> [Detail] 2327 matching rows collapsed into 2229 unique spatial points.

-> [Detail] 4543 empty-region keys cover 24051 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 8.

-> [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22401.60 ms

[+] ZK Prover Time: 1931.81 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.50 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D Parallel M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 42747.79 ms (42.75 s)

Total Client Verification Time: 158.80 ms

3D (50 works)

[*] Global Setup Time: 4603.70 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 4938.02 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543
4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/parallel delegation
5. Prefix Extraction Time: 11324.88 ms
6. Query-Range Key Material: 14.27 KB
7. Global Empty Parent Key Material Touched: 1737.30 KB
8. Query-Scoped Empty Key Material: 1553.73 KB
9. WKD-IBE Delegation Time: 8286.72 ms

-> Engine A Total: 19611.60 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 167.0362 ms (SUCCESS! Complete key cover.)

-> [Detail] 2327 matching rows collapsed into 2229 unique spatial points.

-> [Detail] 4543 empty-region keys cover 24051 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 8.

-> [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 23786.80 ms

[+] ZK Prover Time: 1958.47 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.47 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D Parallel M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 45356.87 ms (45.36 s)

Total Client Verification Time: 169.51 ms

3D (100 works)

[*] Global Setup Time: 4630.90 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 4870.13 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543
4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/parallel delegation
5. Prefix Extraction Time: 11525.31 ms
6. Query-Range Key Material: 14.27 KB
7. Global Empty Parent Key Material Touched: 1737.30 KB
8. Query-Scoped Empty Key Material: 1553.73 KB
9. WKD-IBE Delegation Time: 7586.37 ms

-> Engine A Total: 19111.69 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

[+] Client Empty-Key Check Time: 151.3619 ms (SUCCESS! Complete key cover.)

-> [Detail] 2327 matching rows collapsed into 2229 unique spatial points.

-> [Detail] 4543 empty-region keys cover 24051 empty spatial points.

-> [Detail] Maximum overlap multiplicity among selected cover blocks: 8.

-> [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

[+] ZK Commitment Time: 22212.69 ms

[+] ZK Prover Time: 1934.99 ms

[+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB

[+] Client ZK Verifier Time: 2.45 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D Parallel M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 43259.36 ms (43.26 s)

Total Client Verification Time: 153.81 ms

3D (10 works)

[*] Global Setup Time: 4415.63 ms

[*] Data Path: /home/xing/poneglyphdb/src/data/lineitem_120K.tbl

[*] Row Limit: first 120000 lineitem rows

[*] Query Bounds: shipdate [1994-01-01, 1994-12-31], encoded discount [5, 7] (scale=100), quantity [0, 23]

[*] Loaded 120000 real TPC-H records.

[*] Query matched 2327 real records.

=== DATABASE EMPTY INDEX SETUP ===

[+] Database Empty Index Build Time: 4314.72 ms

[+] Indexed X-Prefix Parent Nodes: 5049

[+] Indexed (X,Y)-Prefix Parent Nodes: 155274

=== UPLOAD STAGE: SKIPPED ===

[*] Use -upload-keys to materialize WKD-IBE keys for every database row.

=== ENGINE A: 3D M-HIBE XY-PARENT SUPPLEMENT (Confidentiality & Access Control) ===

1. Query Canonical Range Keys: 24
2. Global Empty Parent Regions: 20985
3. Query-Scoped Empty Regions: 4543

4. Cover Strategy: offline global (X,Y)-parent empty-Z regions + online query intersection/parallel delegation
 5. Prefix Extraction Time: 10970.91 ms
 6. Query-Range Key Material: 14.27 KB
 7. Global Empty Parent Key Material Touched: 1737.30 KB
 8. Query-Scoped Empty Key Material: 1553.73 KB
 9. WKD-IBE Delegation Time: 11054.02 ms
- > Engine A Total: 22024.93 ms

=== ENGINE A: CLIENT COMPLETENESS CHECK ===

- [+] Client Empty-Key Check Time: 241.4513 ms (SUCCESS! Complete key cover.)
- > [Detail] 2327 matching rows collapsed into 2229 unique spatial points.
 - > [Detail] 4543 empty-region keys cover 24051 empty spatial points.
 - > [Detail] Maximum overlap multiplicity among selected cover blocks: 8.
 - > [Detail] 64 sampled verification keys successfully decrypted same-pattern WKD-IBE ciphertexts.

=== ENGINE B: ZK-ACCUMULATOR (Authenticity) ===

- [+] ZK Commitment Time: 22635.67 ms
- [+] ZK Prover Time: 2027.74 ms
- [+] ZK Proof Size: 0.91 KB (Mem) + 0.29 KB (Deg) = 1.20 KB
- [+] Client ZK Verifier Time: 2.35 ms (SUCCESS!)

=== ULTIMATE ACADEMIC REPORT ===

Architecture: 3D Parallel M-HIBE XY-Parent Supplement Cover + ZK-Accumulator (Lightweight Auth)

Total Server Proving Time: 46688.34 ms (46.69 s)

Total Client Verification Time: 243.80 ms